

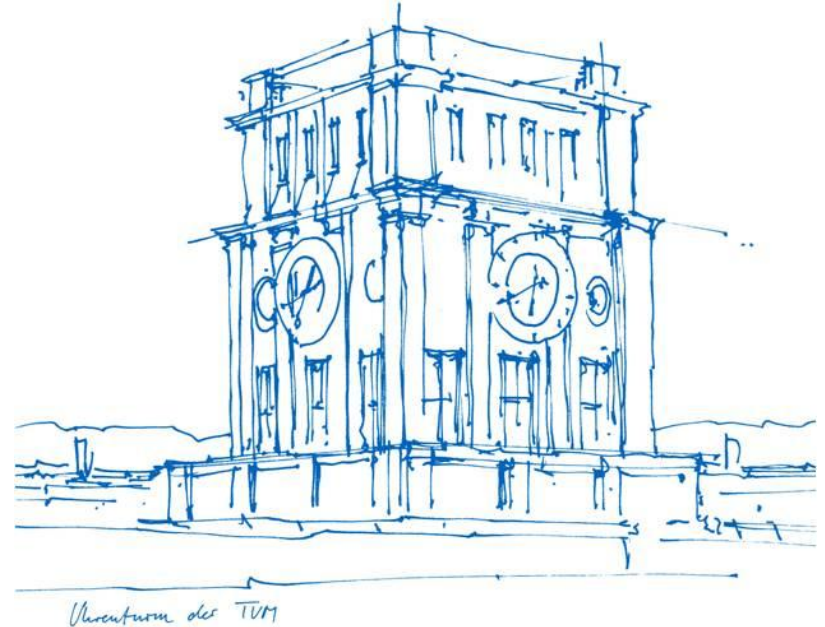
MSMCD Research Lab – Part 1 – AI Accelerator

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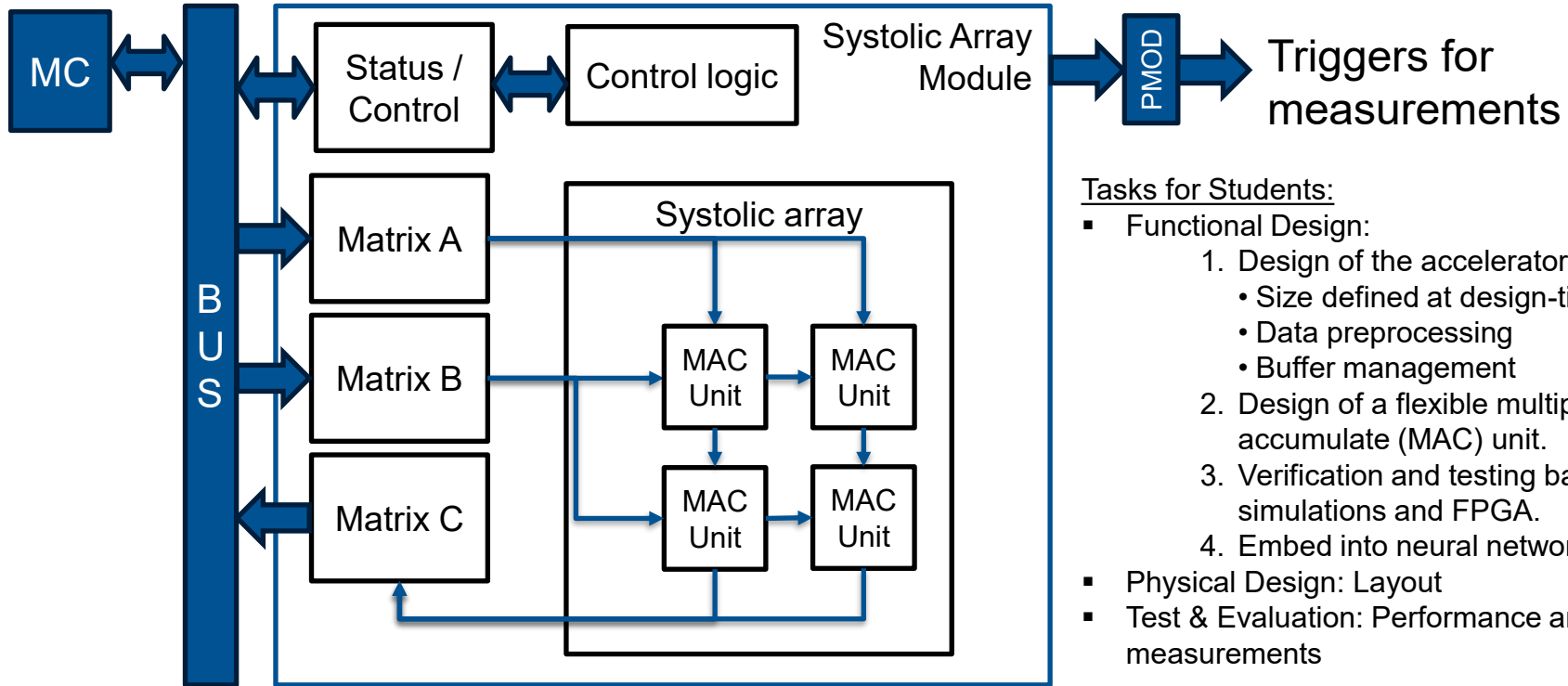
Technical University of Munich

TUM School of Computation, Information and Technology

Chair of AI Processor Design



Project Content



Timeline (Detailed Discussion on Friday)

17 April	24 April
Group forming and introduction.	Block Diagram & Tasks fixed

- Main Tool: Xilinx Vivado
- Main Tool: Synopsys Fusion Compiler

1 May	8 May	15 May	22 May	29 May
Interfaces between blocks agreed and implemented.	RTL for FPGA started, modules have at least dummy behavior.	Tests for RTL code fully prepared.	RTL for FPGA code 90% ready, start of ASIC specific parts.	RTL for FPGA and integration for FPGA completed.

5 June	12 June	19 June	26 June
Design runs on FPGA, Tests on FPGA started.	FPGA Tests completed, ASIC-specific blocks prepared & integrated.	Synthesis for ASIC prepared and running without faults.	Functional tests on gate level successful.

3 July	10 July	17 July
All specifications fulfilled on gate level.	Final deliverable and documentation prepared.	Final deliverable.

We will offer replacement dates for 1 and 8 May;
Regarding final deliverable:
see below.

Milestones (Detailed Discussion on Friday)

1 May: Documentation – Block diagram and interface description.

15 May: Code + Documentation – Tests defined (code for system level can come later).

12 June: Demonstration – Design running on FPGA.

3 July: Code + Documentation: Specification fulfilled for ASIC.

17 July: Code + Documentation: All source files + documentation + presentation.

All RTL code, documentation, etc. must be delivered via a group specific GitLab project (ideally in incremental steps, latest to the dates of the milestones).

FAQs and Remarks

- Additional seminar and shift of dates:
 - The dates on 1 May and 8 May will be shifted (details on Friday).
 - There will be an additional seminar on entrepreneurship → date will be announced soon.
- Attendance for seminars:
 - We expect every week for every group a status update (presentation of < 3 minutes per person+discussion).
 - The default should be: All group members join and present their own work.
- Usage of AI-Tools and sources from the web is allowed but:
 - Has to be declared.
 - You must be able to explain every line of code you have written.
- Improvement of the design over the summer and before the backend part (**first two October weeks!!!**) is possible but without any support from us.
- On Fridays we split the course in two groups:
 - 9:45 – 11:15: groups without export control restrictions.
 - 11:30 – 13:00: groups with export control restriction.

Final Deliverable

- RTL Design
 - Fulfilling the specifications.
 - Passing all test.
 - Clean synthesizable.
 - Delivered via GitLab
- Documentation of design → via GitLab (hint: start from beginning and reuse weekly reporting).
- Final Presentation (decision expected in May):
 - Most likely (planned but not yet confirmed): “Exhibition”, i.e., flash talk + poster presentation + demonstration with industry partners with DECS professors.
 - Backup: Presentation + demonstration.

NDA, License, and Export Control

- Export Control:
 - First check done by TUM based on the declared citizenship.
 - Passport for persons not under export control will be checked when signing the NDA/License Agreement form.
- You have received an email with
 - Link to NDA/license agreements
 - The text of the document to be signed.
- You need to sign the NDA/license agreement document in the technical introduction session.
- Access to tools/PDKs/standard cell libraries will be granted when and for the period needed for the lab.

Group Forming

Members of the same group

- Must:
 - Have the same export control (and NDA) category.
 - Be able to work together efficiently.

- Shall:
 - Have interest in different building blocks.

- Form a group now!
 - Work together today and meet the next few days.
 - In urgent cases we try to exchange participants between groups.
 - We fix the groups on Friday (ideally for 1.5 years).